

NEAT NOW

Project Proposal



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Chapter 1

Neat Now

1.1 Abstract

Effective waste management in housing societies is essential for maintaining cleanliness, ensuring resident satisfaction, and promoting sustainable living environments. Traditional methods, such as manual reporting of waste issues or periodic inspections, often lead to delayed responses and inefficient resource allocation. This project proposes an innovative AI-driven waste management system tailored for housing societies, utilizing a self-built custom computer vision model to automate the detection and classification of waste from user-uploaded images. The system incorporates three primary roles: residents (users), who upload images of waste; an admin, who reviews AI-generated insights; and a cleaning team, responsible for waste removal. By leveraging advanced computer vision techniques, the system identifies waste and estimates quantities, mapping these to specific locations within the society for rapid intervention. A web-based dashboard provides real-time visualization of waste accumulation, enabling efficient task allocation and reducing reliance on manual processes. This solution aims to enhance cleanliness, streamline operations, and foster sustainable waste management practices within housing societies, ultimately contributing to healthier and more eco-friendly communities.

1.2 Introduction

Waste management in housing societies is critical for maintaining hygienic, aesthetically pleasing, and sustainable living spaces. Effective waste management ensures resident well-being, prevents environmental degradation, and enhances the quality of community life. However, many housing societies rely heavily on reactive methods, such as resident complaints or scheduled manual inspections, which

on their own often result in delayed responses and inefficient use of resources. While manual inspections remain important, these traditional approaches fail to provide real-time insights into waste accumulation, leading to prolonged periods of neglect in common areas such as parks, pathways, or waste collection points.

To address these challenges, this project proposes an innovative, AI-powered waste management system designed specifically for housing societies. The system leverages a self-built custom computer vision model to analyze images uploaded by residents, automatically detecting waste and estimating waste quantities. The system integrates three key roles: residents, who capture and upload images of waste; an admin, who receives and reviews AI-generated insights; and a cleaning team, tasked with addressing waste accumulation. By mapping waste-prone areas within the society and tracking the duration of uncleaned zones, the system enables proactive and data-driven decision-making. A web-based dashboard, built using modern technologies, provides visualization of waste data, empowering society management to prioritize cleaning efforts efficiently. This project aims to revolutionize waste management in housing societies, ensuring cleaner, healthier, and more sustainable community environments.

1.3 Problem Statement

Housing societies struggle with poor waste segregation, inefficient collection, and lack of monitoring, causing health risks and pollution. An AI-driven system is urgently needed to automate detection, streamline management, and create cleaner, sustainable communities.

1.4 Objectives

The objectives of this project are as follows:

- **Develop an AI-powered system** that uses a self-built custom computer vision model to detect the waste and estimate quantities from images uploaded by residents in a housing society.
- **Enable waste detection** by analyzing user-uploaded images and highlighting waste-prone areas on a map of the housing society, marking them in red or orange for quick identification.
- **Track waste accumulation duration**, measuring how long specific areas remain unclean to support timely intervention and data-driven decision-making.

- **Create an interactive dashboard** for society management to visualize waste data, enabling efficient task allocation to cleaning teams.
- **Integrate automated and manual waste identification and management**, combining AI-powered detection with scheduled inspections to optimize waste management operations within the housing society.

1.5 Related Work

Waste management systems in housing societies often rely on manual reporting or basic technological interventions, such as bin-level sensors. However, these systems lack area-wide waste detection capabilities. The following related systems highlight current approaches and their limitations:

- **Manual Complaint Systems [1]:** Many housing societies depend on resident complaints via phone calls or emails, which are reactive and lack automated detection, leading to delays in addressing waste issues.
- **Smart Bin Monitoring Systems [5]:** Systems like Sensoneo use sensors to monitor bin fill levels, optimizing waste collection. However, they focus solely on bins and do not address litter in open areas such as parks or pathways within a housing society.
- **ConvoWaste [6]:** This system focuses on post-collection waste segregation using AI but does not provide waste detection accumulation in public areas, limiting its applicability to proactive management.
- **Labellerr AI [8]:** While capable of detecting waste in images, it lacks spatial and temporal tracking, which is critical for prioritizing cleaning efforts in a housing society context.
- **Revolutionizing Urban Solid Waste Management with AI and IoT [9]:** This review explores AI and IoT integration for urban waste, but it emphasizes large-scale city applications rather than tailored solutions for smaller communities like housing societies.
- **Smart Waste Management: A Paradigm Shift Enabled by Artificial Intelligence [10]:** Discusses AI for real-time monitoring in smart bins, yet overlooks user-driven image uploads and community-specific mapping.

- **Multifamily Smart Waste Management Solution by RealPage [11]:** Uses cameras and AI for waste monitoring in multifamily housing, but focuses primarily on cost reduction through bin optimization without comprehensive geospatial tracking.
- **Implementing AI-Driven Waste Management Systems in Underserved Communities [12]:** Provides a framework for AI in low-income areas, highlighting scalability, but lacks emphasis on resident engagement via mobile uploads.
- **Artificial Intelligence Based System for Sorting and Detection of Waste [13]:** Employs AI for waste sorting, demonstrating viability, but is limited to post-collection processes and does not integrate real-time community reporting.
- **AI-Driven Innovations in Waste Management: Catalyzing the Circular Economy [14]:** Investigates AI for circular economy at various levels, but does not address housing society-specific challenges like limited space and resident awareness.
- **Artificial Intelligence for Waste Management in Smart Cities [15]:** Reviews AI for intelligent bin systems to enhance efficiency and reduce diseases, but is geared toward city-scale rather than localized housing environments.

TABLE 1.1: Related Systems Comparison

System/Study	Weakness	Proposed Solution
Manual Complaints [1]	Reactive, no real-time detection.	Automated waste detection and mapping using a self-built model for proactive management.
Smart Bins [5]	Limited to bin-level monitoring, ignores open-area litter.	Custom computer vision model detects waste in open areas via resident-uploaded images.
ConvoWaste [6]	Focuses on post-collection, not real-time monitoring.	Real-time waste detection and classification in society areas pre-collection.

Labellerr AI [8]	Lacks spatial/temporal tracking.	Integrates geospatial mapping and duration tracking for prioritized cleaning.
AI and IoT in Urban Waste [9]	Emphasizes city-scale, not community-tailored.	Focuses on housing society with user uploads and custom model for localized detection.
Smart Waste Paradigm Shift [10]	Overlooks user-driven inputs.	Incorporates resident image uploads for community engagement.
RealPage Multifamily Solution [11]	Prioritizes cost over comprehensive tracking.	Adds geospatial and time-based analysis for efficient interventions.
AI in Underserved Communities [12]	Lacks resident engagement focus.	Empowers users with image uploads and AI insights.
AI Waste Sorting System [13]	Post-collection only.	Enables pre-collection detection in real-time.
AI for Circular Economy [14]	Broad levels, misses society specifics.	Tailored for housing with custom model and dashboard.
AI in Smart Cities Waste [15]	City-oriented, not localized.	Adapts AI for housing societies with mapping and roles.

1.6 Proposed Methodology/System

To address inefficient waste management in housing societies, this project proposes an AI-based system using a self-built custom computer vision model for automated waste detection and management. The methodology includes the following key steps:

- Data Collection:** A curated dataset will be created to train the computer vision model for waste detection in the housing society. The training data will combine (a) public labeled urban-waste image datasets for general pre-training, and (b) a custom dataset collected manually by the project team across the society's common areas (parks, pathways, bin areas, and walkways). Team-collected images will be labeled and marked with bounding boxes or segmentation masks as needed. To make the model robust, the

dataset will include images taken under different lighting, with occlusion, and with varied backgrounds

- **Waste Detection and Analysis:** The self-built custom computer vision model processes uploaded images to detect waste and estimate its quantity by AI model. OpenCV will be used for image processing, and TensorFlow will support model development and training. A heatmap of waste accumulation will be generated to visually identify the most problematic areas within the society, enabling targeted interventions.
- **Geospatial Mapping:** Detected waste areas are mapped to specific locations within the housing society using predefined coordinates or GPS data from user devices. A map-based interface, built with Leaflet.js, highlights waste-prone areas in red or orange and cleaner areas in green, integrating the heatmap for enhanced visualization.
- **Time-based Analysis:** The system tracks how long waste remains un-addressed in specific areas using timestamped image data. Pandas will be used for time-series analysis, enabling prioritization of cleaning tasks based on duration of neglect. A monthly clean society leaderboard will be generated from this data, encouraging eco-friendly behavior by ranking areas or residents based on cleanliness contributions.
- **Dashboard Visualization:** A web-based dashboard, developed using Django and React.js, provides immediate waste reports uploaded by society residents to admin. The dashboard displays an interactive map of the society, quantity insights, and task assignments for the cleaning team. Plotly and Leaflet.js will enhance data visualizations and map interactions. Data analytics features, including charts, cleaning efficiency, and trends, will be integrated to support decision-making. While prior studies focus on city-level infrastructure or post-collection processing, our system addresses the micro-level needs of housing societies. It is lightweight, scalable, and designed for communities with limited budgets but high expectations for cleanliness.
- **Mobile App Development:** A dedicated mobile app will be developed for both residents and cleaning staff, facilitating faster reporting of waste issues by residents and providing immediate updates and task assignments for cleaning teams. The app will include English Language support to cater the housing societies..

- **Task Automation and Notifications:** Task automation will auto-assign cleaning tasks to teams based on waste location and workload, optimizing resource allocation. Awareness notifications will be sent to residents via the mobile app, reminding them about waste segregation and recycling practices to promote sustainable behavior. Current solutions stop at detection but do not ensure accountability in cleanup. Our system directly integrates with cleaning teams by allocating tasks and enabling completion verification, ensuring a closed feedback loop.

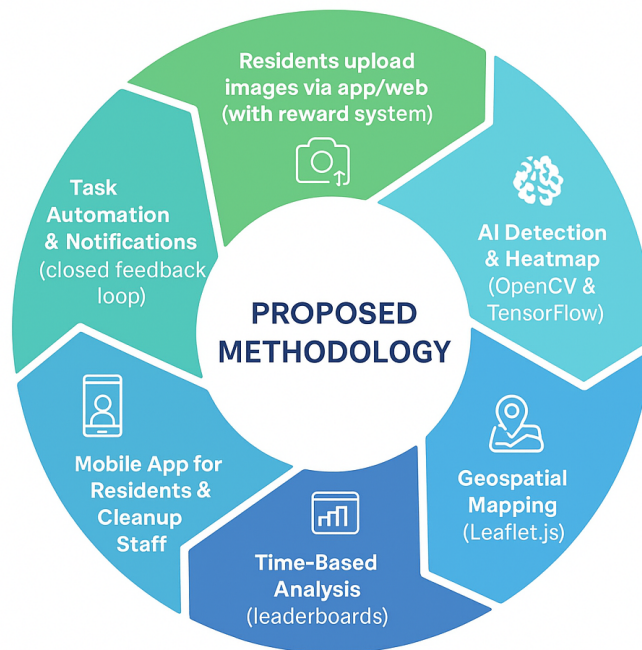


FIGURE 1.1: Methodology Diagram

1.7 Tools and Technologies

- **Programming Language: Python (Version 3.x):** Used for image processing, machine learning, data analysis, and web development.
- **Computer Vision and Image Processing: OpenCV (Version 4.5.x):** For processing resident-uploaded images and detecting waste.
- **Machine Learning Framework:**
 - **TensorFlow (Version 2.x):** Used to develop and train a self-built custom computer vision model for waste detection and classification.

- **Object Detection Approach:** To enable automated monitoring, object detection technique will be employed using a custom-built model to identify waste and debris from images, balancing speed and accuracy for reliable performance.
- **Geospatial Mapping & Visualization: Leaflet.js (Version 1.x):** For displaying interactive maps of the housing society, highlighting waste-prone areas.
- **Web Framework: Django (Version 3.x):** For developing the back-end of the dashboard, handling data storage, and providing API support.
- **Frontend Development: React.js (Version 17.x):** For building an interactive, user-friendly dashboard interface.
- **Data Analysis: Pandas and Numpy:** For time-series analysis and tracking waste accumulation duration.
- **Data Visualization: Plotly:** For creating interactive visualizations of waste trends and metrics.

1.8 Team Members Individual Tasks/Work Division

Team Member	Tasks
Alisha	• Front-end development using React.js • Designing the resident app and admin dashboard interface • Integrating Leaflet.js for interactive map visualization
Qasim	• Backend development using Django • API development for user uploads and admin-cleaning team communication • Data handling and storage
Noor-ul-Ain Ilyas	• Development and training of self-built custom computer vision model using TensorFlow

1.9 Data Gathering Approach

The project will rely on a combination of publicly available datasets and a custom dataset tailored to the housing society context. Open-source datasets containing

Team Member	Tasks
Noor-ul-Ain Ilyas	• Training and validating waste detection models • Testing model performance on society-specific datasets
Ahmad	• Image processing and waste detection using OpenCV • waste analysis system integration • Geospatial mapping with Leaflet.js

TABLE 1.2: Team Members and Their Assigned Tasks

labeled images of urban waste will be used to pre-train the self-built custom computer vision model. To ensure context-specific accuracy, a custom dataset will be created by collecting images from the housing society, either through resident uploads or manual capture by the team. These images will be labeled and used to train the model. If access to resident-uploaded images is limited initially, the team will simulate data collection by capturing images of common areas within the society, ensuring the model is trained on relevant and diverse data.

1.10 Timeline/Gantt Chart

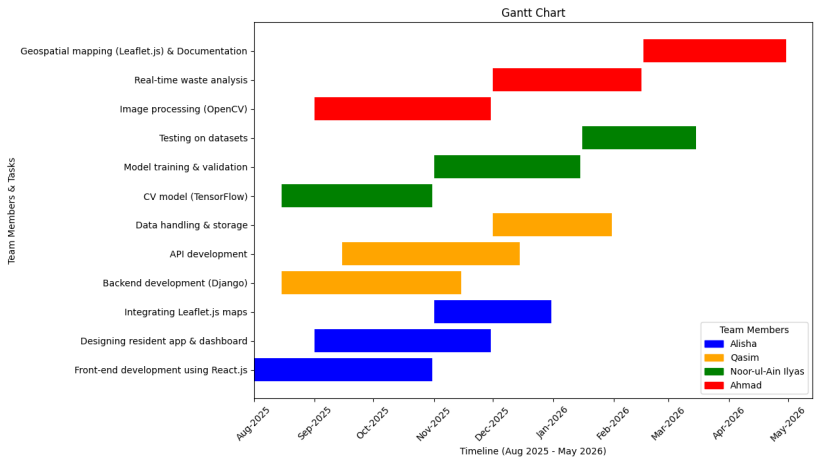


FIGURE 1.2: Gantt Chart

1.11 References

1. Lahore Waste Management Company. “Lahore Waste Management Company Official Website.” 2025. LWMC. Accessed April 2025. <https://www.lwmc.com.pk/>.
2. Sensoneo. “Smart Waste Monitoring System.” 2025. Sensoneo. Accessed April 2025. <https://sensoneo.com/product/ultrasonic-bin-sensors/>.

3. Nafiz, Md. Shahariar, et al. "ConvoWaste: An Automatic Waste Segregation Machine Using Deep Learning." arXiv, February 2023. <https://doi.org/10.48550/arXiv.2302.02976>.
4. Labellerr. "Vision AI Use Cases in Waste Detection." 2025. Labellerr. Accessed April 2025. <https://labellerr.com/>.
5. Abdallah, M., et al. "Revolutionizing urban solid waste management with AI and IoT: A comprehensive review." Environmental Advances, 2025. <https://www.sciencedirect.com/science/article/pii/S2590123025001069>.
6. Abdallah, M., et al. "Smart waste management: A paradigm shift enabled by artificial intelligence." Artificial Intelligence in the Life Sciences, 2024. <https://www.sciencedirect.com/science/article/pii/S2949750724000385>.
7. RealPage. "Multifamily Smart Waste Management Solution." 2025. Accessed August 2025. <https://www.realpage.com/waste-management-solution/>.
8. Oluwaseun, A. "Implementing AI-Driven Waste Management Systems in Underserved Communities in the USA." ResearchGate, March 2024. https://www.researchgate.net/publication/379040258_IMPLEMENTING_AI-DRIVEN_WASTE_MANAGEMENT_SYSTEMS_IN_UNDERSERVED_COMMUNITIES_IN_THE_USA.
9. Thesai.org. "Artificial Intelligence Based System for Sorting and Detection of Waste." International Journal of Artificial Intelligence and Applications, May 2025. https://thesai.org/Downloads/Volume16No5/Paper_9-Artificial_Intelligence_Based_System.pdf.
10. Ramachandran, S., et al. "AI-Driven Innovations in Waste Management: Catalyzing the Circular Economy." Conference Proceedings of the 5th International Conference on the Future of Waste Management, June 2025. <https://www.mdpi.com/2673-4591/97/1/12>.
11. Abdallah, M., et al. "Artificial intelligence for waste management in smart cities: a review." Environmental Chemistry Letters, May 2023. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10169138/>.